Cloud Computing – Assignment 2

Subject: Cloud Computing
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1. Evolution from Classic Data Center to Virtualization and Cloud Computing

Traditional data centers rely on physical servers hosted on■premises, where each application runs on a separate machine. This results in high capital expenditure, poor resource utilization, and complex maintenance. Scaling infrastructure requires purchasing new hardware, which is time■consuming and costly.

Virtualization introduced hypervisors that allow multiple virtual machines to run on a single physical server. Each VM operates independently with its own operating system. This improves hardware utilization, reduces energy consumption, and enables faster deployment.

Cloud computing evolved from virtualization by delivering IT resources as services over the internet. Users can access infrastructure, platforms, and software on demand. Cloud computing provides scalability, elasticity, high availability, and a pay■as■you■go pricing model. Organizations move to cloud platforms to reduce costs, improve agility, and support global operations.

2. Comparison of AWS, Microsoft Azure, and Google Cloud Platform (GCP)

AWS, Azure, and GCP are leading cloud providers. AWS offers the widest range of services and has the largest global infrastructure. It is suitable for startups and large enterprises.

Microsoft Azure integrates well with Microsoft technologies such as Windows Server, Active Directory, and Office 365. It is popular among enterprises and supports hybrid cloud environments.

Google Cloud Platform is known for its strength in big data, analytics, and machine learning. Services like BigQuery and TensorFlow make it ideal for data■driven applications. Pricing models also differ among providers.

3. Amazon Web Services (AWS) Overview

Amazon Web Services is a comprehensive cloud computing platform that provides computing, storage, databases, networking, and security services. AWS allows organizations to deploy applications without managing physical infrastructure.

AWS operates globally through regions, availability zones, and edge locations. Regions are geographical areas, availability zones are isolated data centers, and edge locations deliver content with low latency. AWS ensures high availability through data replication and fault■tolerant architecture.

4. Ways to Access AWS Services

AWS services can be accessed using the AWS Management Console, AWS Command Line Interface (CLI), and AWS Software Development Kits (SDKs). The console provides a graphical interface, CLI enables automation, and SDKs allow developers to integrate AWS services into applications.

5. AWS Security Management

AWS follows a shared responsibility model. IAM allows administrators to manage users, roles, and permissions. STS provides temporary credentials to improve security.

Resource Access Manager enables sharing of resources across accounts. Single Sign■On provides centralized authentication, and Amazon Cognito manages user authentication for web and mobile applications. Best practices include using MFA, least privilege access, and continuous monitoring.

6. AWS Security and Encryption Services

AWS provides services such as KMS for encryption key management, CloudHSM for hardware security, Shield for DDoS protection, and WAF for web application security.

Amazon GuardDuty detects threats, while CloudWatch and CloudTrail monitor and log activities. Billing alerts help manage cloud costs effectively.

7. Traditional Storage vs Cloud Storage

Traditional storage systems have limited scalability and require high maintenance. Backup and disaster recovery are complex.

Cloud storage offers scalability, durability, and global accessibility. Amazon S3 is an object■based storage service with extremely high durability and availability.

8. Comparison of Amazon S3, EBS, and EFS

Amazon S3 is used for object storage such as backups and static content. Amazon EBS provides block storage for EC2 instances and is commonly used for operating systems and databases. Amazon EFS provides a shared file system accessible by multiple EC2 instances.

9. Amazon EC2 Architecture

Amazon EC2 provides virtual servers using AMIs. Key pairs ensure secure access, security groups act as firewalls, and VPC provides network isolation.

Elastic Network Interfaces manage networking, IP addressing enables communication, instance store provides temporary storage, and snapshots are used for backups.

10. AWS Block and File Storage Services

Amazon EBS provides persistent block storage, EFS provides scalable file storage, and Amazon FSx supports Windows and high■performance workloads. The choice depends on performance, scalability, and application requirements.